Developer PowerShell

```
** Visual Studio 2026 Developer PowerShell v18.4.2
** Copyright (c) 2026 Microsoft Corporation
**********************************************************************
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D> deno test gcd_test.ts
Warning Implicitly using latest version (0.224.0) for https://deno.land/std/testing/asserts.ts
Check gcd_test.ts
running 1 test from ./gcd_test.ts
gcdBruteForce: ggT of 1 and 1 should be 1 ... FAILED (3ms)

 ERRORS 

gcdBruteForce: ggT of 1 and 1 should be 1 => ./gcd_test.ts:4:6
error: AssertionError: Values are not equal.


    [Diff] Actual / Expected


-   0
+   1

  throw new AssertionError(message);
        ^
    at Module.assertEquals (https://deno.land/std@0.224.0/assert/assert_equals.ts:52:9)
    at assertEquals (https://deno.land/std@0.224.0/testing/asserts.ts:101:11)
    at file:///C:/Users/atieh/OneDrive%20-%20sluz/Dokumente/Lehre/Freitag/1Jahre-%202%20Semester/M426-Freitag/A
ufgaben/6-aufgabe-TDD/gcd_test.ts:6:3

 FAILURES 

gcdBruteForce: ggT of 1 and 1 should be 1 => ./gcd_test.ts:4:6

FAILED | 0 passed | 1 failed (32ms)

error: Test failed
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
```

gcd_test.ts* | gcd.ts

```
1
2  import { assertEquals } from "https://deno.land/std/testing/asserts.ts";
3  import { gcdBruteForce } from "./gcd.ts";
4
5  Deno.test("gcdBruteForce: ggT of 1 and 1 should be 1", () => {
6    const result = gcdBruteForce(1, 1);
7    assertEquals(result, 1);
8  });
```

118 % | Keine Probleme gefunden | Zeile: 8, Zeichen: 4 | TABULATOREN | CRLF | UTF-8

gcd_test.ts* | gcd.ts

```
1  export function gcdBruteForce(a: number, b: number): number {
2    return 0;
3  }
```

118 % | Keine Probleme gefunden | Zeile: 1, Zeichen: 1 | TABULATOREN | CRLF | UTF-8

118 % Keine Probleme gefunden Zeile: 4, Zeichen: 2 TABULATOREN CRLF UTF-8

Developer PowerShell

```
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D> deno test gcd_test.ts
Check gcd_test.ts
running 1 test from ./gcd_test.ts
gcdBruteForce: ggT of 1 and 1 should be 1 ... ok (1ms)

ok | 1 passed | 0 failed (14ms)

PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D>
```

Developer PowerShell Fehlerliste Ausgabe

gcd_test.ts gcd.ts

```typescript
export function gcdBruteForce(a: number, b: number): number {
  if (a === 1 && b === 1) return 1;
  return 0;
}
```

gcd_test.ts gcd.ts

```typescript

import { assertEquals } from "https://deno.land/std/testing/asserts.ts";
import { gcdBruteForce } from "./gcd.ts";

Deno.test("gcdBruteForce: ggT of 1 and 1 should be 1", () => {
  const result = gcdBruteForce(1, 1);
  assertEquals(result, 1);
});
```

Developer PowerShell

```
ok | 1 passed | 0 failed (14ms)

PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D> deno test gcd_test.ts
Check gcd_test.ts
running 1 test from ./gcd_test.ts
gcdBruteForce: Table-Driven Tests ... FAILED (3ms)

ERRORS

gcdBruteForce: Table-Driven Tests => ./gcd_test.ts:10:6
error: AssertionError: Values are not equal: Failed for gcd(6, 9)


    [Diff] Actual / Expected


-   0
+   3

  throw new AssertionError(message);
        ^
    at Module.assertEquals (https://deno.land/std@0.224.0/assert/assert_equals.ts:52:9)
    at assertEquals (https://deno.land/std@0.224.0/testing/asserts.ts:101:11)
    at file:///C:/Users/atieh/OneDrive%20-%20sluz/Dokumente/Lehre/Freitag/1Jahre-%202%20Semester/M426-Freitag/A
ufgaben/6-aufgabe-TDD/gcd_test.ts:13:5

FAILURES

gcdBruteForce: Table-Driven Tests => ./gcd_test.ts:10:6

FAILED | 0 passed | 1 failed (30ms)

error: Test failed
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D>
```

Developer PowerShell Fehlerliste Ausgabe

gcd_test.ts | gcd.ts

```typescript
import { assertEquals } from "https://deno.land/std/testing/asserts.ts";
import { gcdBruteForce } from "./gcd.ts";

const gcdTests = [
  { a: 1, b: 1, expected: 1 },
  { a: 6, b: 9, expected: 3 },
  { a: 81, b: 36, expected: 9 }
];

Deno.test("gcdBruteForce: Table-Driven Tests", () => {
  for (const { a, b, expected } of gcdTests) {
    const actual = gcdBruteForce(a, b);
    assertEquals(actual, expected, `Failed for gcd(${a}, ${b})`);
  }
});
```

118 % | Keine Probleme gefunden | Zeile: 3, Zeichen: 1 | SPC | CRLF | UTF-8

gcd_test.ts | gcd.ts

```typescript
export function gcdBruteForce(a: number, b: number): number {
  if (a === 1 && b === 1) return 1;
  return 0;
}
```

Developer PowerShell

```
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D> deno test gcd_test.ts
Check gcd_test.ts
running 1 test from ./gcd_test.ts
gcdBruteForce: Table-Driven Tests ... ok (0ms)

ok | 1 passed | 0 failed (8ms)

PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D>
```

Developer PowerShell | Fehlerliste | Ausgabe

gcd_test.ts | gcd.ts

```typescript
export function gcdBruteForce(a: number, b: number): number {
  let start = Math.min(a, b);

  for (let i = start; i >= 1; i--) {
    if (a % i === 0 && b % i === 0) {
      return i;
    }
  }
  return 1;
}
```

gcd_test.ts | gcd.ts

```typescript
import { assertEquals } from "https://deno.land/std/testing/asserts.ts";
import { gcdBruteForce } from "./gcd.ts";

const gcdTests = [
  { a: 1, b: 1, expected: 1 },
  { a: 6, b: 9, expected: 3 },
  { a: 81, b: 36, expected: 9 }
];

Deno.test("gcdBruteForce: Table-Driven Tests", () => {
  for (const { a, b, expected } of gcdTests) {
    const actual = gcdBruteForce(a, b);
    assertEquals(actual, expected, `Failed for gcd(${a}, ${b})`);
  }
});
```

Developer PowerShell

```
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D> deno test fraction_test.ts
Check fraction_test.ts
running 1 test from ./fraction_test.ts
Fraction.cancel: 18/27 should become 2/3 ... FAILED (2ms)

 ERRORS 

Fraction.cancel: 18/27 should become 2/3 => ./fraction_test.ts:4:6
error: AssertionError: Values are not equal.


    [Diff] Actual / Expected


-   18
+   2

  throw new AssertionError(message);
        ^
    at Module.assertEquals (https://deno.land/std@0.224.0/assert/assert_equals.ts:52:9)
    at assertEquals (https://deno.land/std@0.224.0/testing/asserts.ts:101:11)
    at file:///C:/Users/atieh/OneDrive%20-%20sluz/Dokumente/Lehre/Freitag/1Jahre-%202%20Semester/M426-Freitag/A
ufgaben/6-aufgabe-TDD/fraction_test.ts:7:3

 FAILURES 

Fraction.cancel: 18/27 should become 2/3 => ./fraction_test.ts:4:6

FAILED | 0 passed | 1 failed (16ms)

error: Test failed
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D>
```

Developer PowerShell | Fehlerliste | Ausgabe

fraction.ts | fraction_test.ts | gcd_test.ts | gcd.ts

```typescript
import { gcdBruteForce } from "./gcd.ts";

export class Fraction {
  constructor(public numerator: number, public denominator: number) {}

  cancel(): void {
  }
}
```

fraction.ts | fraction_test.ts | gcd_test.ts | gcd.ts

```typescript
import { assertEquals } from "https://deno.land/std/testing/asserts.ts";
import { Fraction } from "./fraction.ts";

Deno.test("Fraction.cancel: 18/27 should become 2/3", () => {
  const f = new Fraction(18, 27);
  f.cancel();
  assertEquals(f.numerator, 2);
  assertEquals(f.denominator, 3);
});
```

Developer PowerShell | Fehlerliste | Ausgabe

```
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D> deno test fraction_test.ts
Check fraction_test.ts
running 1 test from ./fraction_test.ts
Fraction.cancel: 18/27 should become 2/3 ... ok (0ms)

ok | 1 passed | 0 failed (7ms)

PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D>
```

fraction.ts | fraction_test.ts | gcd_test.ts | gcd.ts

```typescript
import { gcdBruteForce } from "./gcd.ts";

export class Fraction {
  constructor(public numerator: number, public denominator: number) {}

  cancel(): void {
    const commonFactor = gcdBruteForce(this.numerator, this.denominator);
    this.numerator = this.numerator / commonFactor;
    this.denominator = this.denominator / commonFactor;
  }
}
```

fraction.ts | fraction_test.ts | gcd_test.ts | gcd.ts

```typescript
import { assertEquals } from "https://deno.land/std/testing/asserts.ts";
import { Fraction } from "./fraction.ts";

Deno.test("Fraction.cancel: 18/27 should become 2/3", () => {
  const f = new Fraction(18, 27);
  f.cancel();
  assertEquals(f.numerator, 2);
  assertEquals(f.denominator, 3);
});
```

fraction.ts | fraction_test.ts | gcd_test.ts* | gcd.ts*

```typescript
export function gcdBruteForce(a: number, b: number): number {
  let start = Math.min(a, b);

  for (let i = start; i >= 1; i--) {
    if (a % i === 0 && b % i === 0) {
      return i;
    }
  }
  return 1;
}

export function gcdEuclid(a: number, b: number): number {
  if (a === b) return a;
  if (a > b) return gcdEuclid(a - b, b);
  return gcdEuclid(a, b - a);
}
```

fraction.ts | fraction_test.ts | gcd_test.ts* | gcd.ts*

```typescript
import { assertEquals } from "https://deno.land/std/testing/asserts.ts";
import { gcdBruteForce } from "./gcd.ts";

const gcdTests = [
  { a: 1, b: 1, expected: 1 },
  { a: 6, b: 9, expected: 3 },
  { a: 81, b: 36, expected: 9 }
];

/*Deno.test("gcdBruteForce: Table-Driven Tests", () => {
  for (const { a, b, expected } of gcdTests) {
    const actual = gcdBruteForce(a, b);
    assertEquals(actual, expected, `Failed for gcd(${a}, ${b})`);
  }
});*/

Deno.test("gcdEuclid: Table-Driven Tests", () => {
  for (const { a, b, expected } of gcdTests) {
    const actual = gcdEuclid(a, b);
    assertEquals(actual, expected);
  }
});
```

Developer PowerShell | Fehlerliste | Ausgabe

```
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D> deno test fraction_test.ts
Check fraction_test.ts
running 1 test from ./fraction_test.ts
Fraction.cancel: 18/27 should become 2/3 ... ok (0ms)

ok | 1 passed | 0 failed (7ms)

PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D> deno test gcd_test.ts
running 1 test from ./gcd_test.ts
gcdBruteForce: Table-Driven Tests ... ok (1ms)

ok | 1 passed | 0 failed (31ms)

PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
```

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```
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D> deno test fraction_test.ts
Check fraction_test.ts
running 1 test from ./fraction_test.ts
Fraction: automatic cancellation in constructor ... FAILED (2ms)

ERRORS

Fraction: automatic cancellation in constructor => ./fraction_test.ts:11:6
error: AssertionError: Values are not equal.


    [Diff] Actual / Expected


-   10
+   1

  throw new AssertionError(message);
        ^
    at Module.assertEquals (https://deno.land/std@0.224.0/assert/assert_equals.ts:52:9)
    at assertEquals (https://deno.land/std@0.224.0/testing/asserts.ts:101:11)
    at file:///C:/Users/atieh/OneDrive%20-%20sluz/Dokumente/Lehre/Freitag/1Jahre-%202%20Semester/M426-Freitag/A
ufgaben/6-aufgabe-TDD/fraction_test.ts:13:3

FAILURES

Fraction: automatic cancellation in constructor => ./fraction_test.ts:11:6

FAILED | 0 passed | 1 failed (21ms)

error: Test failed
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D>
```

fraction.ts | fraction_test.ts | gcd_test.ts | gcd.ts

```typescript
import { gcdBruteForce } from "./gcd.ts";

export class Fraction {
  constructor(public numerator: number, public denominator: number) {}

  cancel(): void {
    const commonFactor = gcdBruteForce(this.numerator, this.denominator);
    this.numerator = this.numerator / commonFactor;
    this.denominator = this.denominator / commonFactor;
  }
}
```

fraction.ts | fraction_test.ts | gcd_test.ts | gcd.ts

```typescript
import { assertEquals } from "https://deno.land/std/testing/asserts.ts";
import { Fraction } from "./fraction.ts";

/*Deno.test("Fraction.cancel: 18/27 should become 2/3", () => {
  const f = new Fraction(18, 27);
  f.cancel();
  assertEquals(f.numerator, 2);
  assertEquals(f.denominator, 3);
});*/

Deno.test("Fraction: automatic cancellation in constructor", () => {
  const f = new Fraction(10, 20);
  assertEquals(f.numerator, 1);
  assertEquals(f.denominator, 2);
});
```

Developer PowerShell | Fehlerliste | Ausgabe

Developer PowerShell

```
PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D> deno test fraction_test.ts
Check fraction_test.ts
running 1 test from ./fraction_test.ts
Fraction: automatic cancellation in constructor ... ok (1ms)

ok | 1 passed | 0 failed (10ms)

PS C:\Users\atieh\OneDrive - sluz\Dokumente\Lehre\Freitag\1Jahre- 2 Semester\M426-Freitag\Aufgaben\6-aufgabe-TD
D>
```

fraction.ts | fraction_test.ts | gcd_test.ts | gcd.ts

```typescript
import { gcdBruteForce } from "./gcd.ts";

export class Fraction {
  constructor(public numerator: number, public denominator: number) {
    this.cancel();
  }
  cancel(): void {
    const commonFactor = gcdBruteForce(this.numerator, this.denominator);
    this.numerator = this.numerator / commonFactor;
    this.denominator = this.denominator / commonFactor;
  }
}
```

fraction.ts | fraction_test.ts | gcd_test.ts | gcd.ts

```typescript
import { assertEquals } from "https://deno.land/std/testing/asserts.ts";
import { Fraction } from "./fraction.ts";

/*Deno.test("Fraction.cancel: 18/27 should become 2/3", () => {
  const f = new Fraction(18, 27);
  f.cancel();
  assertEquals(f.numerator, 2);
  assertEquals(f.denominator, 3);
});*/

Deno.test("Fraction: automatic cancellation in constructor", () => {
  const f = new Fraction(10, 20);
  assertEquals(f.numerator, 1);
  assertEquals(f.denominator, 2);
});
```